

Exercise 3: Sorting Customer Orders

CODE:

```
class Order {
    int orderId;
    String customerName;
    double totalPrice;
    Order(int orderId, String customerName, double totalPrice) {
        this.orderId = orderId;
        this.customerName = customerName;
        this.totalPrice = totalPrice;
    }
    void display() {
        System.out.println(orderId + "\t" + customerName + "\t\t" + totalPrice);
    }
}

public class Main {
    // Bubble Sort
    static void bubbleSort(Order[] orders) {
        int n = orders.length;
        for (int i = 0; i < n - 1; i++) {
            for (int j = 0; j < n - i - 1; j++) {
                if (orders[j].totalPrice > orders[j + 1].totalPrice) {
                    Order temp = orders[j];
                    orders[j] = orders[j + 1];
                    orders[j + 1] = temp;
                }
            }
        }
    }
}
```

```

// Quick Sort

static void quickSort(Order[] orders, int low, int high) {
    if (low < high) {
        int pivot = partition(orders, low, high);
        quickSort(orders, low, pivot - 1);
        quickSort(orders, pivot + 1, high);
    }
}

static int partition(Order[] orders, int low, int high) {
    double pivot = orders[high].totalPrice;
    int i = low - 1;
    for (int j = low; j < high; j++) {
        if (orders[j].totalPrice <= pivot) {
            i++;
            Order temp = orders[i];
            orders[i] = orders[j];
            orders[j] = temp;
        }
    }
    Order temp = orders[i + 1];
    orders[i + 1] = orders[high];
    orders[high] = temp;
    return i + 1;
}

static void displayOrders(Order[] orders) {
    System.out.println("Order ID\tCustomer\tPrice");
    for (Order order : orders) {
        order.display();
    }
}

```

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    }
}

public static void main(String[] args) {
    Order[] bubbleOrders = {
        new Order(101, "Aarav", 4500),
        new Order(102, "Diya", 2200),
        new Order(103, "Rahul", 7800),
        new Order(104, "Sneha", 3100)
    };

    Order[] quickOrders = {
        new Order(101, "Aarav", 4500),
        new Order(102, "Diya", 2200),
        new Order(103, "Rahul", 7800),
        new Order(104, "Sneha", 3100)
    };

    bubbleSort(bubbleOrders);

    System.out.println("Bubble Sort Result");

    displayOrders(bubbleOrders);

    System.out.println();

    quickSort(quickOrders, 0, quickOrders.length - 1);

    System.out.println("Quick Sort Result");

    displayOrders(quickOrders);
}
}

```

OUTPUT:

Bubble Sort Result		
Order ID	Customer	Price
102	Diya	2200.0
104	Sneha	3100.0
101	Aarav	4500.0
103	Rahul	7800.0

Quick Sort Result		
Order ID	Customer	Price
102	Diya	2200.0
104	Sneha	3100.0
101	Aarav	4500.0
103	Rahul	7800.0